

MATTHEW GONG

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in matthew-x-gong



mattxgong

AWARDS

- **American Invitational Math Exam 2022** (Advanced from American Math Competition 12)
- **Canadian Computing Competition 2021**: Top 7%
- **USA Computing Olympiad Bronze 2020**: 1000 (Perfect Score)
- **University of Waterloo**: President's Scholarship of Distinction and CFM Entrance Award

SKILLS

- **Programming**: Python, C++, SQL, Javascript, TypeScript, Java, Swift, CSS, HTML, C, R#
- **Libraries/Frameworks**: PyTorch, Google Agent Development Kit, Multi-Agent Systems, Hugging Face, TensorFlow, Node.js, Puppeteer, React.js, pandas
- **Tools/Platforms**: Git, Linux, ROS, Google AI Studio, IOS, Power BI, Power Apps, Power Automate, AWS, Docker, SAP, Anaconda, Jupyter Notebook, .NET, Xcode, Unity

EDUCATION

Bachelor in Computing and Financial Management, *University of Waterloo* 2023–2027 (Expected)

INTERESTS

- Ultimate Frisbee, Tennis, Ski, Snowboarding, Swimming, Guitar, Piano, Escape Rooms

EXPERIENCE

TECHNICAL ANALYST, MPBSDP

Apr 2026 – Oct 2026

- ◇ Led a full frontend migration from Angular 4 to Angular 20, using a multi-agent RPI workflow, producing concrete artifacts at each phase. Enforced continuous validation loops through automated Karma and Jasmine test suites, executing iterative refinement cycles to eliminate regressions and deliver a resilient, maintainable, future-proof application.
- ◇ Architected advanced Multi-Agent-and-Skill Systems, leveraging AI-driven generation of PowerShell and Python automation scripts to interface with external developer platforms including Azure DevOps and Figma. Applied Context and Harness engineering techniques to orchestrate automated task planning, implementation of tasks, streamline PR code reviews, and dynamically generate comprehensive test suites.
- ◇ Developed autonomous multi-agent pipelines to construct and maintain Karpathy-style LLM Wikis, bypassing traditional vector store infrastructure for personal knowledge bases. Integrated hybrid RAG architecture to enable seamless information retrieval, allowing deep context retention and emergent insights to compound over time.

DATA ANALYST, Ontario Financing Authority (OFA)

Jan 2025 – Aug 2025

- ◇ Collaborated with data team to produce detailed analysis of Interest-on-Debt, the Government of Ontario's 5th largest expense, with **Python**, **pyODBC**, **SQL**, and **Git** on both a monthly and daily basis, organizing and categorizing over **\$1 billion** in expenditures.
- ◇ Developed an automatic resume filtering system using **Python**, **Regular Expression (Regex)**, and **NLP** techniques to parse, analyze, and rank resumes based on job-specific keywords and qualifications, streamlining the manual review of over 1000 resumes and enhancing productivity.

FINANCE INTERN, Trench Group (Siemens Energy)

May 2024 – Aug 2024

- ◇ Took the initiative to design and implement a **Power Apps** tool to automate corporate expense reporting, set up an email alert and approval process via **Power Automate**, created visual and interactive analysis on expenses trend and breakdowns via **Power BI**.
- ◇ Used **Regex**, **OCR with Google's Tesseract** and PDF libraries in Python to identify over **3 million USD** discrepancies by conducting extensive comparisons between the production backlog and purchase orders.

PROJECTS

ROBOTICS CHALLENGE

July 2025 – August 2025

- ◇ Competed in an **Embodied Artificial Intelligence** and **robotics** team challenge in China, retrained a You Only Look Once (YOLO) computer vision neural network to achieve real-time image segmentation and object detection; Used **Python**, **Robot Operating System (ROS) libraries** to program autonomous navigation with stair climbing, picking up objects, turning and walking for a humanoid robot in a simulated environment. Won **3rd-Place** Award among hundreds of university teams across China.

ROS MOBILE CONTROLLER

Sept 2023 – April 2024

- ◇ An iOS app that remotely communicates with a robot running **ROS** to maneuver the robot via a simulated joystick. It creates and displays a real-time map of the robot's surrounding environment using **C++** and **Simultaneous Localization and Mapping (SLAM)** techniques. Optimized message handling and network throughput to ensure stable control and reliable state estimation in dynamic environments. Managed sensor data streams to support robust state estimation and spatial awareness.

